

- Introspection - William James (1800s) [Subjective]

- Psychology

↳ Study of "Psyche"
↳ Mind

Experimental Psychology

Social Psychology - How groups behave

Physiological Psychology - What brain areas are important for some behaviour

Cognitive Psychology - How knowledge influences behaviour

↳ To think and know knowledge

Disadvantages of Introspection:

Inconsistent

Individualistic - Vary from person to person

∴ Next Method: Behaviourism (Objective) - 1900s - Watson

Basis for cognition: Behaviour

Stimulus (Input) $\xrightarrow{\text{Association}}$ Response (Output)

(i) Avoidance Behaviour } Response
(ii) Approach Behaviour }

Stimuli can be any input

Cognition
(Genes) / (Multiple Stimuli ↔ Response)
wired learnt

Cognition

How do we know
what we know

Repeated Association → Habit

There is nothing called mind

Pass the Salt

Sass the Palt

Salt, Please! (Meaning mediating)

Could you pass the salt?

Next Method: Cognitive Revolution (1950-60's)

- Cognitivism

- Tolman, Miller

Capacity limitations [7 ± 2 chunks]
(Working memory)

Temporary buffer

- Chomsky

Associations doesn't explain how children learn language

Universal Grammar (UG)

- Counterfactual reasoning

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→ Cocktail Party Problem $\Rightarrow$ Forceful Attention

→ Consciousness - (i) Alive, Awake and Ability to respond to Stimuli

(ii) Ability to experience and reflect on experiences (Polia)

Self-Conscious $\Rightarrow$ Humans

4th Epoch: Conscious

→ Brain - basis of behaviour:

Neural Basis for Cognition

Spirit Basis?

Avg. Response time: 100 ms

Temporal Lobe: Language

How do we study cognition?

Behaviour

- Cognitive Psychology

- Experiments

(BRED course)

$\hookrightarrow$ (BRSM course)

Brain

- Cognitive Neuroscience

}

Healthy

Non-Invasive Invasive
(fMRI, MEG, EEG) (PET, ECoG)

Unhealthy

① Accidents
Brain Injury, Disease

- Broca's Aphasia

(Expressive Speech Disorder) - Frontal Lobe

- Wernicke's Aphasia

(Comprehensive Speech Disorder) - Temporal Lobe

• Capgras Syndrome

- Recognition

- Memory

- Comparison

- Emotion Evaluation

Integration

Amygdala: Fearful Stimuli

I - Behaviour

Brain: Product of Evaluation

II - Brain

III - Computational Model

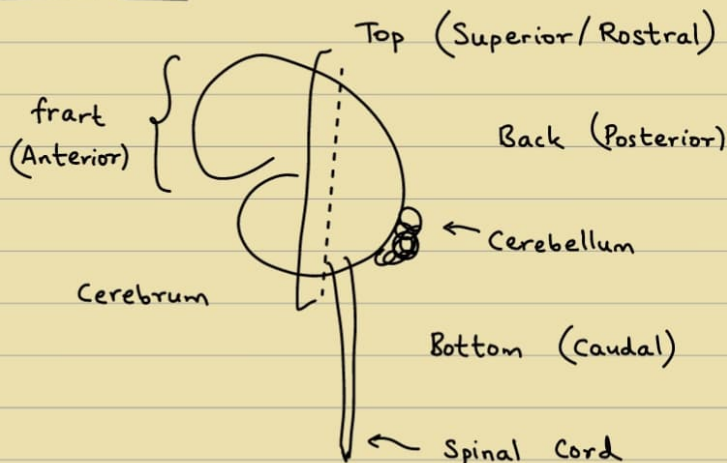
Mathematical Model ↔ Experiment

[INCM]

Complex functionalities

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→ Neural Basis:

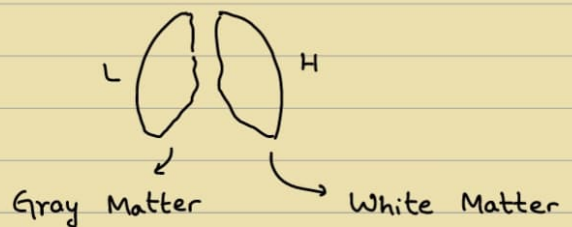


Meninges

Pia Matter

Two Hemispheres:

LH and RH



Coronal Section

Vertebral bones protect Spinal Cord.

Fluid surrounds this is Cerebrospinal Fluid (CSF)

- Skull Bone

- Central Nervous System :

Cerebellum + Spinal Cord

Receives processes and gives out output

- Peripheral Nervous System :

Generates signal for muscles to contract

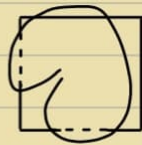
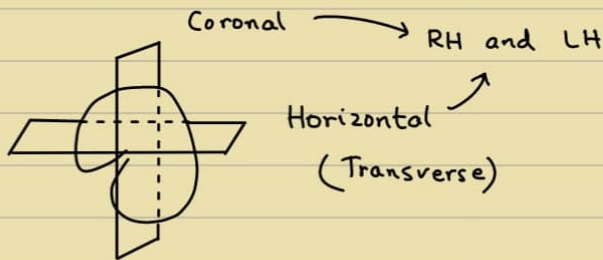
- Dopamine : Pleasure

Gray Matter : Neural Cell body

White Matter : Fibres / Axons

(White coating) that connect

• Sections of the brain :

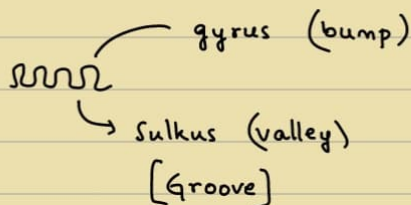


(Mid-sagittal)

Ventral : Front

Dorsal : Back

Cortex :



Big Grooves are called Fissures

Central Sulcus : Separates front and back

Lateral Fissure : Separates top and bottom

Horizontal Fissure : Separates left and right (Longitudinal)

Transverse Fissure : Separates Cerebrum and Cerebellum

- 4 Lobes :

- (1) Frontal Lobe :

Cerebrum : Neocortex

- Well developed
 - Newest part of Neocortex
 - Keeps developing until 21 years
 - Working Memory , Executive function
 - Planning
 - Emotional Evaluation (Phineas Gage) , Personality
 - Decision Making
 - Motor Cortex
 - Speech (Brocas Area)

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(Out-of-Sight , Out-of-mind)

- (2) Paretal Lobe :

- Somatosensory cortex
 - Proprioception : Sense of where their body parts are
 - Goal-directed Action
 - (Spatial Orientation)

- CNS and PNS work together

- (3) Occipital Lobe :

- Visual Cortex : Vision
 - Object Recognition , Object Identification
 - Color , Shape , Size
 - Object Motion (Movement of Object)

- (4) Temporal Lobe :

- Object Naming
 - Language
 - Speech Comprehension (Verniki's Area)
 - Auditory Processing (Auditory Cortex)
 - Input from ears

- Subcortical Struct

- Amygdala (Fear)
 - Hippocampus (Long-Term Memory)
 - Basal Ganglia (Motor)
 - Thalamus (Gathers incoming things and distribute to appropriate)
 - (i.e. Sensory Exchange - Waystation for Sensory

- Cerebellum
 - Motor Coordination
 - Compromised for Long-Term Alcoholics
- Hypothalamus (Mid-Brain / Old-brain)
 - Hunger
 - Breathing control / Respiration control
 - Temp. Control
 - Sleep - Wake Cycle

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- Dementia
- Functional MRI
- Coma : Vegetative State

- Reptilian Complex



Limbic system : Happy, sad, etc



Neocortex

- EEG : Electrode Cap
 - Deep Sleep : Low Frequency, High Altitude
 - Fully Awake : High Frequency, Low Altitude

- fMRI : Response ↓ when visualising

Blood Oxygenation / Deoxygenation

Brain Computer Interface

Avg. Connectivity in Neural System : 10^4

→ Deep Brain Simulation - Parkinson's Disease
Action Potential - 10^{-3} s

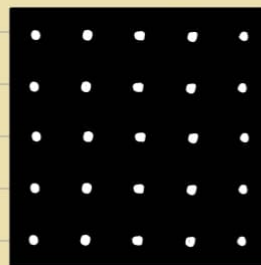
Nature & Nurture

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→ Perception : Interpretation of raw input data

- Neurons - Electrically excitable
Axon - Carry information
- Synapses - Electrochemical Majorly
Gap Junction - Reduced Evolutionarily

Visual Illusion :



- Hodgkin and Huxley - Squid

Resting : -70 V

Threshold : -55 V

- Inter - Pulse Intervals

- Action Potentials to Perception :
Modularity

- Grandmother Coding - Localist theory

i.e. one neuron corresponds to face of Grandmother

Multimodal distribution - Distributed Processing

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Thought Process : Reasoning and Decision Making

- Human : Sequential (Speech) &
Parallel (Speech + Vision)

Attention - Sustained Attention
- Divided Attention

Desire gives ability to Plan, Execute and Fulfills

- Emotion : Sense of Morality, Empathy

Perception : Interpretation

- Top down : Allows you to determine what is unclear / Ambiguous
Bottom Up : See the illusion directly
- Short Term Memory - Hours (1 hr, 2 hrs)
Procedural Memory : Motor Memory
Priming $\equiv$ Memory
 $\hookrightarrow$ Prepotentiation of our system so that in future,
 we can see patterns quickly
- Homunculus : Declarative Memory
 Coronal Section

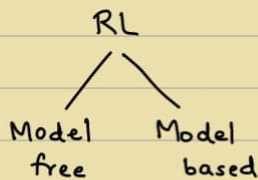
Fine grip - Not in animals, only in Humans

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Primary & Secondary Mortar Cortex - Feedback Policy (Frontal)

State Estimation - Parietal lobe

Frontal lobe & Basal Ganglia - Cost and Reward

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- BRED

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PreFrontal lobe : Higher Order Thinking (Complex Tasks)

Expectation of the task, Stores & generates expectation
Stores meanings

Sensory Memory → Short-term Memory → Long-term Memory
[Attention] (Temporal Lobe) [Hippocampus]
(Occipital lobe)

- Short - Term Memory :

Chunking : 7 ± 2 chunks

- Long - Term Memory :

Perma Memory

- Working Memory

Digit Span

Operation Span - Working Memory Capacity

Reading Span

- WM Model :

Dreaming - Visuospatial Sketchpad

Concurrent Articulation : Talking to Inner self

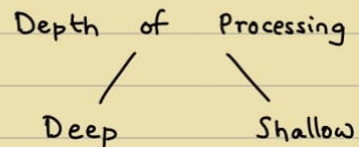
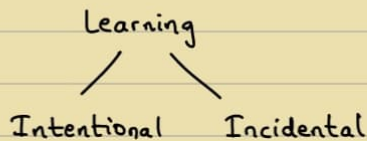
Episodic Buffer : Connects everything together

Missing in Amnesia Patients

- Recency Effect : Distraction can be used to reduce effect

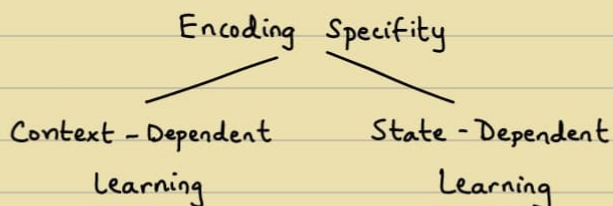
Primacy Effect : ?

- Learning :

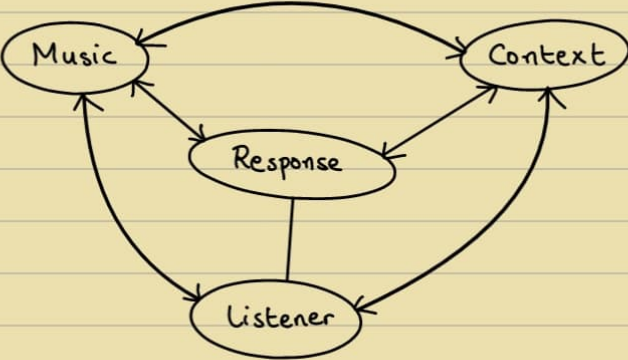


- Context :

Memory stores context



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Concentration
Focus

→ Attention:

- Selectivity
Inferial Temporal Cortex : Naming of Object, Labeling of Object
- Activity:

X	M	L	T
A	F	N	B
P	C	W	A

J	T	L	Q
D	G	I	Q
Y	W	C	S